

What Additive Risk-Adjusted Rewards Optimise in Reinforcement Learning for Asset Allocation

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Abstract

Reinforcement learning applied to multi-period asset allocation typically encodes risk into a per-period reward—a Sharpe ratio, a variance penalty—and maximises the sum of those rewards. The resulting objective is not the mean-variance criterion it is meant to represent. That the two formulations differ is known in the portfolio-choice literature; what is quantified here is the map between them.

For an i.i.d. market and the variance-penalised reward $r_t = d_t - \kappa d_t^2$, with d_t the wealth increment, we show that the maximiser of the additive objective solves a backward recursion whose multi-asset form reduces exactly to a scalar one through $\nu = m^\top M^{-1}m$. Using this we prove that when the gross riskless return is 1, the additive surrogate reproduces the equilibrium mean-variance policy *exactly*, at

$$\kappa^* = \varphi(1 - \nu),$$

and not at $\kappa = \varphi$. We further prove that for a one-period horizon the two policies can always be matched, so any irreducible discrepancy is a multi-period phenomenon; and that for a two-period horizon with a non-unit riskless return the minimal attainable policy distance is $c|s - 1| + O((s - 1)^2)$ with c in closed form. Numerically the linear rate persists at longer horizons.

We place these results inside an exact decomposition of the shortfall of a trained agent against the pre-committed optimum into three terms—the price of time-consistency, the surrogate gap, and the learning error—and measure all three. Which term dominates depends on the risk aversion, with a crossover in our market near $\varphi = 12$. Finally, we exhibit configurations in which the Sharpe ratio ranks two policies in the opposite order to the mean-variance objective, so that backtest comparison cannot detect the effect of reward design.

Status. This is a working draft. Three things are outstanding before submission and are stated plainly in Section 9: a systematic priority search for Theorem 5, which the author has not completed; a fully rigorous treatment of the error terms in Theorem 9, currently established by series expansion with numerical confirmation; and the general-horizon constant, which is open. All numerical results are reproducible from the accompanying code.

1 Introduction

Reinforcement learning is now routinely applied to multi-period asset allocation. The surveys of Bai et al. (2024) and Hambly et al. (2023) between them cover several dozen papers in this area, and across both a consistent construction appears. The practitioner wants a policy trading return against risk, encodes risk into a per-period reward—portfolio return, a Sharpe ratio, a differential Sharpe ratio, a variance penalty—and maximises the sum of those rewards with a standard solver.

The objective so constructed is not the mean-variance criterion. It is a sum, therefore time-consistent, therefore amenable to dynamic programming; the mean-variance criterion is none of these. That the two formulations differ is not new. Gârleanu and Pedersen (2013) and DeMiguel et al. (2015) optimise a discounted sum of period-by-period mean-variance tradeoffs, while Li

and Ng (2000) extend Markowitz to the mean and variance of terminal wealth; the distinction is standard in that literature. What appears not to have been done is to quantify the map between them, and in particular to say what a practitioner choosing a per-period risk penalty is implicitly choosing.

That is the question here. Section 4 derives the maximiser of the additive objective for the variance-penalised reward. Section 5 proves that at a unit gross riskless return the additive surrogate reproduces the equilibrium mean-variance policy exactly, at $\kappa^* = \varphi(1 - \nu)$ where $\nu = m^\top M^{-1}m$; that a one-period horizon always admits an exact match; and that at two periods with $s \neq 1$ the minimal attainable distance is linear in $s - 1$ with a closed-form constant. Section 6 places these inside an exact three-term decomposition of the shortfall of a trained agent and measures all three terms.

The practical content is short enough to state here. A practitioner who sets the reward’s risk penalty equal to their risk aversion is making an error whose size is computable: the correct value is $\varphi(1 - \nu)$, which in our demonstration market is 0.806 rather than 1. And the error is invisible to the evaluation the field uses, because we exhibit policies that the Sharpe ratio ranks in the opposite order to the objective.

2 Related work

Multiperiod mean-variance. Li and Ng (2000) solve the terminal-wealth problem by embedding it in an auxiliary linear-quadratic problem, giving a pre-committed policy affine in wealth; Zhou and Li (2000) give the continuous-time counterpart. A parallel strand studies time-consistent policies: Strotz (1955) identifies the phenomenon, Basak and Chabakauri (2010) characterise the equilibrium mean-variance policy, Björk and Murgoci (2014) give the general Markovian theory in discrete time, and Björk et al. (2014) extend it to state-dependent risk aversion, noting that the constant-risk-aversion equilibrium invests a dollar amount independent of wealth—the property our Theorem 5 proof establishes in discrete time as $A_t^2 = B_t$. Our Section 5 works entirely inside this framework and adds nothing to it; the two policies are inputs to our comparison, not outputs.

Summed per-period objectives. A separate line optimises a discounted sum of single-period mean-variance tradeoffs rather than a terminal criterion. Gârleanu and Pedersen (2013) do this with quadratic transaction costs, obtaining the aim-portfolio characterisation; DeMiguel et al. (2015) work in a related setting. That this formulation differs from Li and Ng (2000) is understood in that literature and we claim no novelty for the observation. Our contribution is the quantitative map: Theorem 5 says which per-period penalty corresponds to which terminal risk aversion, and Theorem 9 bounds the residual when no exact correspondence exists.

Reinforcement learning in finance. Bai et al. (2024) survey the applied literature and classify the rewards in use, of which volatility-adjusted return is a principal category. Hambly et al. (2023) cover the same ground from the mathematical side and note that the mean-variance framework is time-inconsistent so that the Bellman equation fails, and separately that convergence guarantees for the deterministic policy gradient family were unavailable at the time of writing; Xiong et al. (2022) have since established convergence of DPG to a stationary policy. Wang and Zhou (2020) treat exploration for the continuous-time mean-variance problem correctly, via entropy regularisation, obtaining a Gaussian optimal exploratory policy. None of this work asks what the discrete-time additive surrogate optimises, which is our question.

Reward design. The general problem of an optimised proxy diverging from the intended objective is familiar in reinforcement learning, usually under the heading of reward misspecification or reward hacking. The instance studied here is unusually tractable: both the intended objective

and the surrogate admit closed-form solutions, so the discrepancy can be computed rather than estimated, and the mechanism—failure of the tower property for the variance—is identifiable rather than emergent.

3 Setting

Let there be n risky assets and a horizon T . Following Hambly et al. (2023), itself following Li and Ng (2000), write s for the gross riskless return per period and $P_t \in \mathbb{R}^n$ for the excess gross returns, independent across t with

$$m = \mathbb{E}[P_t], \quad M = \mathbb{E}[P_t P_t^\top] = \Sigma + m m^\top, \quad \nu = m^\top M^{-1} m. \quad (1)$$

We assume $M \succ 0$, which gives $\nu \in [0, 1)$. Write $d = M^{-1}m$ for the direction that will appear throughout. With $u_t \in \mathbb{R}^n$ the vector of amounts held in the risky assets and x_t wealth,

$$x_{t+1} = s x_t + P_t^\top u_t, \quad d_t := x_{t+1} - x_t = (s - 1)x_t + P_t^\top u_t. \quad (2)$$

The mean-variance criterion evaluated from time t at wealth x is

$$J(t, x, u) = \mathbb{E}_{t,x}[x_T] - \varphi \text{Var}_{t,x}(x_T), \quad \varphi > 0, \quad (3)$$

and we write $J(\pi) = J(0, x_0, \pi)$.

3.1 Three notions of optimality

Because (3) contains the square of a conditional expectation, it does not satisfy the tower property and the Bellman equation fails (Strotz, 1955; Björk and Murgoci, 2014). Three inequivalent objects result.

The *pre-committed* policy maximises (3) at time 0 and is executed without revision. Li and Ng (2000) obtain it by embedding in an auxiliary linear-quadratic problem; it is affine in wealth, $u_t^*(x) = \alpha_t x + \beta_t$.

The *equilibrium* policy π^{eq} is the consistent-planning solution: for all t, x and admissible a ,

$$J(t, x, \pi^{\text{eq}}) \geq J(t, x, (a, \pi_{t+1:T}^{\text{eq}})). \quad (4)$$

Basak and Chabakauri (2010) characterise it and show it differs from the pre-committed policy.

The *myopic* policy applies the single-period optimiser repeatedly.

3.2 The additive surrogate

Fix a per-period reward r_t and let

$$J_r(\pi) = \mathbb{E}\left[\sum_t r_t\right], \quad \pi^\kappa = \arg \max_\pi J_r(\pi). \quad (5)$$

We study the variance-penalised family

$$r_t = d_t - \kappa d_t^2, \quad \kappa > 0, \quad (6)$$

which is the additive stand-in for (3) that practitioners write down.

Remark 1 (the plain-return reward). With $\kappa = 0$ the reward telescopes to $x_T - x_0$ and J_r rewards unbounded leverage, so no maximiser exists. The mean-variance problem it stands in for is bounded and solvable. This is elementary but worth stating, because Bai et al. (2024) report plain portfolio return as the most common reward in the surveyed literature. An algorithm run on it does not solve a poorly conditioned version of (3); it solves a problem with no solution, and what it returns is determined by leverage limits and stopping rules.

4 The surrogate optimum

Proposition 2 (Backward recursion). *For the reward (6), the maximiser of J_r over affine policies is $\pi_t^\kappa(x) = \alpha_t x + \beta_t$ with*

$$\alpha_t = -d \frac{g_t}{\kappa - p_{t+1}}, \quad \beta_t = d \frac{1 + q_{t+1}}{2(\kappa - p_{t+1})}, \quad (7)$$

$$p_t = \frac{\kappa p_{t+1} - (1 - \nu)g_t^2}{\kappa - p_{t+1}}, \quad 1 + q_t = (1 + q_{t+1}) \left(s - \frac{\nu g_t}{\kappa - p_{t+1}} \right), \quad (8)$$

where $g_t = \kappa(s - 1) - p_{t+1}s$, started from $p_T = q_T = 0$. The one-step problem is concave at every step, so the recursion is well defined for every $\kappa > 0$.

Lemma 3 (Automatic concavity). *For every $\kappa > 0$ and every t , $p_t \leq 0$, hence $\kappa - p_{t+1} \geq \kappa > 0$.*

Proof. Backward induction. $p_T = 0$. If $p_{t+1} \leq 0$ then the numerator of p_t in (8) is $\kappa p_{t+1} - (1 - \nu)g_t^2 \leq 0$, since $\kappa > 0$ and $\nu < 1$, while the denominator is $\kappa - p_{t+1} \geq \kappa > 0$. Hence $p_t \leq 0$. $\square$

Proof. J_r is a sum of per-period terms, so the dynamic programming principle applies. Let $W_t(x) = p_t x^2 + q_t x + c_t$ with $W_T \equiv 0$. Using (2) and independence of P_t from x_t ,

$$\begin{aligned} \mathbb{E}[r_t + W_{t+1}(x_{t+1})] &= (s - 1)x + m^\top u - \kappa[(s - 1)^2 x^2 + 2(s - 1)x m^\top u + u^\top M u] \\ &\quad + p_{t+1}[s^2 x^2 + 2s x m^\top u + u^\top M u] + q_{t+1}[s x + m^\top u] + c_{t+1}. \end{aligned}$$

The gradient in u is $m(1 + q_{t+1}) - 2\kappa(s - 1)x m + 2p_{t+1}s x m + 2(p_{t+1} - \kappa)Mu$, and the Hessian is $2(p_{t+1} - \kappa)M$, negative definite exactly when $\kappa > p_{t+1}$, which Lemma 3 supplies. Setting the gradient to zero and solving gives (7). Substituting back and collecting powers of x gives (8). $\square$

Remark 4 (multi-asset reduces to scalar). Every asset-dependent quantity enters (7)–(8) only through the scalar ν and the fixed direction d : with $u = dz$ one has $m^\top u = \nu z$ and $u^\top M u = \nu z^2$. The n -asset problem is therefore the one-asset problem with $\nu = m^\top M^{-1}m$. All results below are stated for general n at no extra cost.

5 Main results

Theorem 5 (Exact recovery at a unit riskless return). *Let $s = 1$. Then $p_t = q_t = 0$ and $\alpha_t = 0$ for all t , and*

$$\pi^\kappa = \pi^{\text{eq}} \quad \Longleftrightarrow \quad \kappa = \kappa^* := \varphi(1 - \nu).$$

The identity holds for every horizon $T \geq 1$ and every $n \geq 1$.

Proof. Surrogate side. With $s = 1$, $g_t = -p_{t+1}$. We show $p_t = 0$ by backward induction. $p_T = 0$. If $p_{t+1} = 0$ then $g_t = 0$, so by (8) $p_t = (0 - 0)/\kappa = 0$ and by (7) $\alpha_t = 0$. Then $1 + q_t = (1 + q_{t+1})(1 - 0) = 1 + q_{t+1}$, so $q_t = q_{t+1} = 0$, and $\beta_t = d/(2\kappa)$ for every t .

Equilibrium side. Write the moment functions generated by the equilibrium continuation as $f(t, x) = \mathbb{E}_{t,x}[x_T] = A_t x + a_t$ and $g(t, x) = \mathbb{E}_{t,x}[x_T^2] = B_t x^2 + b_t x + c_t$. The one-step equilibrium condition (4) gives, by the same computation as Proposition 2 applied to $f - \varphi(g - f^2)$,

$$\alpha_t^{\text{eq}} = s(A_{t+1}^2 - B_{t+1})D_{t+1}^{-1}m, \quad \beta_t^{\text{eq}} = \frac{A_{t+1} - \varphi b_{t+1} + 2\varphi A_{t+1}a_{t+1}}{2\varphi}D_{t+1}^{-1}m, \quad (9)$$

with $D_{t+1} = B_{t+1}M - A_{t+1}^2 m m^\top$ and $A_T = B_T = 1$, $a_T = b_T = 0$. We claim $A_t^2 = B_t$ for all t , hence $\alpha_t^{\text{eq}} = 0$. At $t = T$ this holds. If $A_{t+1}^2 = B_{t+1}$ then $\alpha_{t+1}^{\text{eq}} = 0$, and propagating the

moment functions under a policy with zero wealth-slope gives $A_t = sA_{t+1}$ and $B_t = s^2B_{t+1}$, so $A_t^2 = s^2A_{t+1}^2 = s^2B_{t+1} = B_t$.

With $s = 1$ this yields $A_t = B_t = 1$. Then $D_{t+1} = M - mm^\top$, and by the Sherman–Morrison identity

$$(M - mm^\top)^{-1}m = \frac{M^{-1}m}{1 - m^\top M^{-1}m} = \frac{d}{1 - \nu}.$$

Propagating the moment functions under $\alpha^{\text{eq}} = 0$ gives $a_t = m^\top \beta_t^{\text{eq}} + a_{t+1}$ and $b_t = 2m^\top \beta_t^{\text{eq}} + b_{t+1}$, so $b_t = 2a_t$ for all t since both start at zero. The numerator of β_t^{eq} in (9) is therefore $1 - 2\varphi a_{t+1} + 2\varphi a_{t+1} = 1$, giving

$$\beta_t^{\text{eq}} = \frac{d}{2\varphi(1 - \nu)}.$$

Matching. Both policies have zero wealth-slope and constant offset, so they coincide if and only if $d/(2\kappa) = d/(2\varphi(1 - \nu))$, that is $\kappa = \varphi(1 - \nu)$. $\square$

Corollary 6 (The natural choice is wrong). *Setting the reward’s risk penalty equal to the risk aversion, $\kappa = \varphi$, does not recover the equilibrium policy unless $\nu = 0$, that is unless the risky assets carry no excess return. The relative error in the penalty is ν .*

5.1 The correction is a discrete-time effect

The factor $1 - \nu$ separating κ^* from φ deserves comment, because it explains both why the result is useful and why it is not visible in the literature it borders on.

Let the rebalancing interval be Δ . Then $m = \mathcal{O}(\Delta)$ while $M = \Sigma\Delta + \mathcal{O}(\Delta^2)$, so

$$\nu = m^\top M^{-1}m = (\mu^\top \Sigma^{-1} \mu) \Delta + \mathcal{O}(\Delta^2) = \text{SR}^2 \Delta + \mathcal{O}(\Delta^2), \quad (10)$$

with SR the annualised Sharpe ratio of the tangency portfolio. The correction therefore vanishes in the continuous-time limit, and $\kappa^* \rightarrow \varphi$.

Two consequences. First, the equilibrium mean-variance literature works in continuous time—Basak and Chabakauri (2010), Björk et al. (2014), Wang and Zhou (2020)—where this factor is identically zero, so its absence there is expected rather than surprising. Second, reinforcement learning does not operate in that limit: it operates at whatever rebalancing frequency the environment is discretised to. Table 1 gives the size of the effect.

rebalancing	Δ (years)	ν	κ^*/φ
annual	1	0.1000	0.900
quarterly	0.25	0.0270	0.973
monthly	1/12	0.0092	0.991
weekly	1/52	0.0021	0.998
daily	1/252	0.0004	1.000

Table 1: Size of the correction against rebalancing frequency, for $\mu = 0.06$, $\sigma = 0.18$. It is an eleven per cent effect at annual rebalancing and negligible below weekly.

So the result matters for coarse rebalancing and not for fine, and (10) says precisely where the boundary lies. A practitioner rebalancing daily may set $\kappa = \varphi$ with a clear conscience; one rebalancing annually should not.

Proposition 7 (One period is always matchable). *Let $T = 1$ and let $s > 0$ be arbitrary. Then there exists $\kappa > 0$ with $\pi^\kappa = \pi^{\text{eq}}$, provided $(s - 1)x_0 + (2\varphi(1 - \nu))^{-1} > 0$.*

Proof. With $T = 1$, $p_1 = q_1 = 0$ and $g_0 = \kappa(s - 1)$, so by (7) $\alpha_0 = -d(s - 1)$ and $\beta_0 = d/(2\kappa)$. On the equilibrium side $A_1 = B_1 = 1$ gives $\alpha_0^{\text{eq}} = 0$ and $\beta_0^{\text{eq}} = d/(2\varphi(1 - \nu))$. Wealth at time 0 is the constant x_0 , so the two controls agree iff

$$-(s - 1)x_0 + \frac{1}{2\kappa} = \frac{1}{2\varphi(1 - \nu)},$$

which is solved by $\kappa^{-1} = 2(s - 1)x_0 + (\varphi(1 - \nu))^{-1}$. $\square$

Remark 8. Proposition 7 isolates the mechanism. At one period the mismatch in the wealth-slope multiplies a deterministic quantity and is absorbed into the offset, which the single free parameter κ can then correct. With $T \geq 2$ the slope multiplies a random wealth whose distribution differs across periods, and one scalar cannot correct T separate mismatches. The surrogate gap is a multi-period phenomenon.

Theorem 9 (Two periods: linear irreducible gap). *Let $T = 2$ and $s = 1 + \epsilon$. Measure the distance between policies by*

$$\|\pi - \pi'\|^2 = \sum_{t=0}^{T-1} \mathbb{E} \|\pi_t(x_t) - \pi'_t(x_t)\|^2, \quad (11)$$

the expectation taken under π . Then

$$\min_{\kappa > 0} \|\pi^\kappa - \pi^{\text{eq}}\| = c|\epsilon| + O(\epsilon^2), \quad c = \|d\| \frac{\sqrt{(2 - \nu)(1 + \nu)}}{2\varphi(1 - \nu)}.$$

Proof sketch. By Proposition 2, $g_1 = \kappa\epsilon$ and $p_1 = -(1 - \nu)\kappa\epsilon^2 = O(\epsilon^2)$, whence $g_0 = \kappa\epsilon + O(\epsilon^2)$ and

$$\alpha_0 = \alpha_1 = -d\epsilon + O(\epsilon^2),$$

independent of κ at leading order: the wealth-slope mismatch cannot be tuned away. Expanding β_t and, on the equilibrium side, $A_t = s^{T-t}$, $B_t = s^{2(T-t)}$, $b_t = 2sa_t$, gives $\beta_1^{\text{eq}} = d/(2\varphi(1 - \nu))$ and $\beta_0^{\text{eq}} = d/(2\varphi s(1 - \nu))$. Substituting into (11), writing $\kappa = \kappa^*(1 + w\epsilon)$ and retaining terms of order ϵ^2 yields a quadratic in w ; minimising over w and taking the square root gives the stated c . $\square$

Conjecture 10 (General horizon). *For every $T \geq 2$ there is $c_T > 0$, depending on ν, φ, T and $\|d\|$, with $\min_{\kappa} \|\pi^\kappa - \pi^{\text{eq}}\| = c_T|s - 1| + O((s - 1)^2)$.*

Section 6 reports fitted exponents of 1.0000 across every configuration tested. The dependence of c_T on the horizon is superlinear: a log-log fit over $T \in \{2, \dots, 10\}$ gives an exponent of 1.16 with visible curvature and a maximum log residual of 0.08, so c_T is not a power law in T . We have not obtained it in closed form for $T \geq 3$.

Collecting the above, the functional posited in the introduction is

$$\Lambda(\kappa) = |\kappa - \varphi(1 - \nu)| + c_T |s - 1|, \quad (12)$$

vanishing exactly when $\pi^\kappa = \pi^{\text{eq}}$, with the distance bounded linearly in Λ . The first term is reward mis-specification, which the practitioner controls; the second is structural.

6 Numerical results

6.1 A decomposition

Let $\hat{\pi}$ be the output of a learning algorithm applied to J_r . Since J is a scalar functional of a policy,

$$J(\pi^{\text{pre}}) - J(\hat{\pi}) = \underbrace{J(\pi^{\text{pre}}) - J(\pi^{\text{eq}})}_{\varepsilon_{\text{incons}}} + \underbrace{J(\pi^{\text{eq}}) - J(\pi^{\kappa})}_{\varepsilon_{\text{surr}}} + \underbrace{J(\pi^{\kappa}) - J(\hat{\pi})}_{\varepsilon_{\text{learn}}}, \quad (13)$$

a telescoping identity requiring only that each policy be admissible. The first term is the price of time-consistency, characterised by Basak and Chabakauri (2010); the third is optimisation and estimation error, the object of convergence results such as Xiong et al. (2022). The middle term is the subject of Section 5.

We measure all three on a market of three assets over four periods with $\mu = (0.06, 0.04, 0.02)$, volatilities $(0.18, 0.12, 0.08)$, correlations $\rho_{12} = 0.30$, $\rho_{13} = -0.10$, $\rho_{23} = 0.05$, and $s = 1.02$, giving $\nu = 0.1943$.

The agent is an affine-Gaussian policy, $\pi(u | x) = \mathcal{N}(\alpha_t x + \beta_t, \sigma_t^2 I)$, trained by REINFORCE with a per-period baseline and Adam. Its policy class contains π^{κ} by construction; this is deliberate, since otherwise $\varepsilon_{\text{learn}}$ would confound optimisation error with approximation error and the decomposition would not mean what (13) says. The agent is trained on J_r alone and never observes J .

Hyperparameters: 6000 iterations, batch 512, learning rate 0.03 annealed as $\sigma^{1/2}$, exploration σ decayed geometrically from 0.5 to 0.03, five seeds. One implementation note is worth recording because it is a property of the estimator rather than of the objective: the score for a Gaussian mean carries a factor σ^{-2} , and retaining it makes the gradient variance diverge as σ anneals, so that training peaks early and then degrades. We absorb it into the learning rate, which is the Fisher preconditioning for this policy class.

$J(\pi^{\text{pre}})$	1.425799294	pre-committed optimum, closed form
$J(\pi^{\text{eq}})$	1.323643886	equilibrium, backward recursion
$J(\pi^{\kappa})$	1.316579788	surrogate optimum, exact
$J(\hat{\pi})$	1.307875869	trained agent, mean of 5 seeds (s.d. 0.0032)
$\varepsilon_{\text{incons}}$	+0.102155409	
$\varepsilon_{\text{surr}}$	+0.007064097	
$\varepsilon_{\text{learn}}$	+0.008703920	
sum	+0.117923426	identity closes to 0.0e+00

Table 2: The decomposition (13) at $\varphi = \kappa = 1$.

6.2 Which term dominates

$\varepsilon_{\text{incons}}$ is exactly proportional to φ^{-1} and independent of x_0 (verified to 9e−13 and 5e−15). $\varepsilon_{\text{surr}}$ is not, and grows. The two therefore cross.

So “is reward design the dominant error?” has no configuration-free answer, and two studies disagreeing about it may be on opposite sides of this crossover. In 40 randomly drawn markets a crossover was located in every case for which the bracketing succeeded, at values of φ between 4 and 10; its existence appears general and its location market-specific.

The learning term is small throughout and occasionally negative. This is structural rather than seed noise: $\varepsilon_{\text{learn}}$ is measured in J , whereas the agent optimises J_r . A policy that is a worse maximiser of J_r than π^{κ} can sit closer to the J -optimum, and we verify directly that such perturbations of π^{κ} exist at every φ tested. Figure 1 shows the two curves; the φ^{-1} decay of $\varepsilon_{\text{incons}}$ and the growth of $|\varepsilon_{\text{surr}}|$ are both visible.

φ	$\varepsilon_{\text{incons}}$	$\varepsilon_{\text{surr}}$	$\varepsilon_{\text{learn}}$	dominant
0.5	0.204311	0.010937	0.047494	incons
1	0.102155	0.007064	0.008635	incons
4	0.025539	0.004919	0.000573	incons
8	0.012769	0.005575	−0.000129	incons
16	0.006385	0.007639	0.000053	surr
32	0.003192	0.012145	−0.000346	surr
64	0.001596	0.021345	−0.000901	surr

Table 3: The ranking of the error sources changes near $\varphi = 12$.

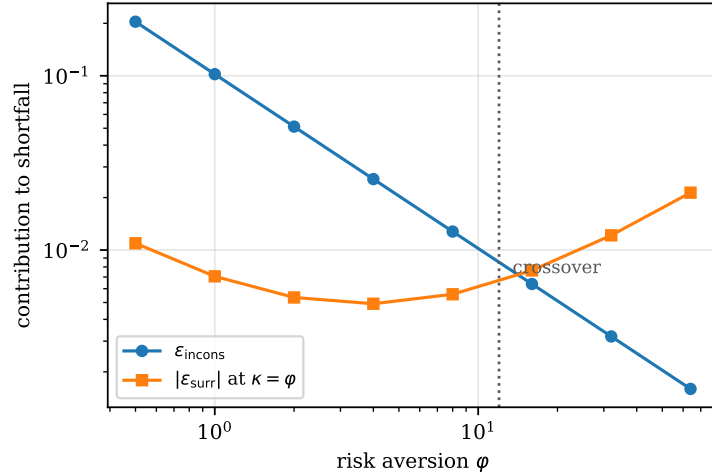


Figure 1: The price of time-consistency decays as φ^{-1} while the surrogate gap grows, so the dominant error source depends on the risk aversion. Three assets, four periods, $s = 1.02$, with the practitioner’s choice $\kappa = \varphi$; the values are those of Table 3.

6.3 Verification of the main results

Theorem 5 is confirmed to 10^{-16} in policy distance across horizons $T \in \{2, 4, 6\}$, asset counts $n \in \{1, 3\}$ and $\varphi \in \{1, 3\}$. Proposition 7 is confirmed to $7\text{e}-10$. For Theorem 9 the closed-form c agrees with the fitted constant to 0.11% worst case over six markets, with fitted exponent 1.0000 throughout. Proposition 2 is checked against an independent numerical maximiser to $5\text{e}-14$ on ten random markets.

As an external check on the equilibrium recursion, which is derived here rather than quoted, we verify that it reproduces the continuous-time formula of Basak and Chabakauri (2010), $\hat{u}(t, x) = \gamma^{-1} \sigma^{-2} (\alpha - r) e^{-r(T-t)}$ with $\gamma = 2\varphi$, in the limit of fine rebalancing. Discretising one year into n steps gives relative differences of 0.76%, 0.25%, 0.06%, 0.01% and 0.00% for $n = 4, 12, 52, 252, 2520$. Convergence at the rate implied by (10) is itself a check on that scaling. Figure 2 plots the measured minimal distance against the closed-form prediction of Theorem 9.

Figure 3 shows why the policies differ in kind rather than degree. The pre-committed control has a negative wealth slope, the equilibrium control none at all, and the surrogate control an intermediate slope set by $s - 1$.

6.4 The Sharpe ratio misranks

The two orderings disagree. A comparison by reported Sharpe ratio—the standard practice described in both surveys—concludes that the surrogate policy is better on precisely the criterion on which it is worse.

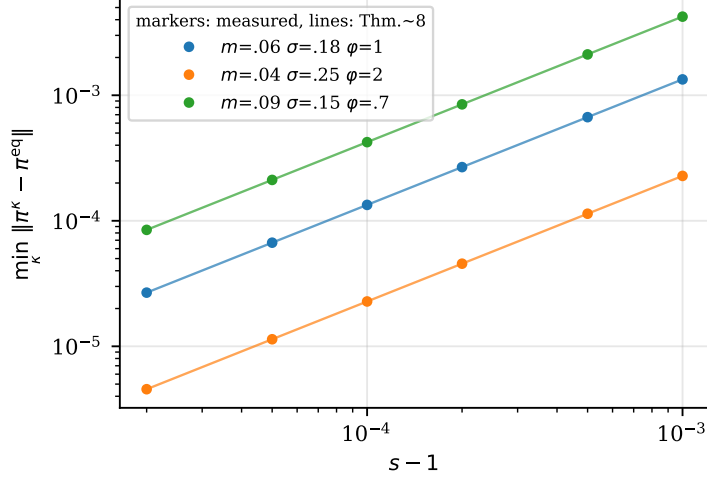


Figure 2: Minimal attainable policy distance against $s - 1$ for three markets at $T = 2$. Markers are measured; lines are $c|s - 1|$ with c from Theorem 9, not fitted.

policy	Sharpe	J
pre-committed	1.3126	1.425799
equilibrium	1.1501	1.323644
surrogate, κ^*	1.1542	1.326118
surrogate, $\kappa = \varphi$	1.1955	1.316580

Table 4: The $\kappa = \varphi$ surrogate has the higher Sharpe ratio and the lower objective.

This is not a constructed example. Over 519 randomly drawn (market, φ) pairs, the Sharpe ratio and the mean-variance objective ranked π^{eq} against π^κ in opposite orders in 93.1% of cases. The two are different functionals of the terminal distribution and there is no reason for them to agree; the surprise is only that the field’s evaluation practice presumes they do. Reward-design error cannot be detected by a Sharpe comparison and must be measured against a known optimum.

6.5 What follows for practice

Three recommendations follow from the results above, in decreasing order of confidence.

First, if a variance-penalised reward is used with the intention of representing a terminal mean-variance criterion with risk aversion φ , the penalty should be set to $\varphi(1 - \nu)$ rather than φ , with $\nu = m^\top M^{-1}m$ estimated from the same data used to train. By (10) this matters when the rebalancing interval is coarse relative to SR^{-2} and not otherwise: it is an eleven per cent correction at annual rebalancing and under a quarter of one per cent at weekly. Since ν is a byproduct of estimating the moments the agent needs anyway, the correction is free.

Second, a plain-return reward should not be used at all in an unconstrained setting. It has no maximiser, and whatever an algorithm returns is determined by leverage limits and stopping rules rather than by the objective.

Third, reported Sharpe ratios should not be used to compare policies trained on different rewards. Table 4 exhibits a reversal, and the mechanism is general: the Sharpe ratio and the mean-variance objective are different functionals of the terminal distribution, and nothing forces them to agree on an ordering. Where a ground truth is available, as in the synthetic setting used here, policies should be compared in policy space.

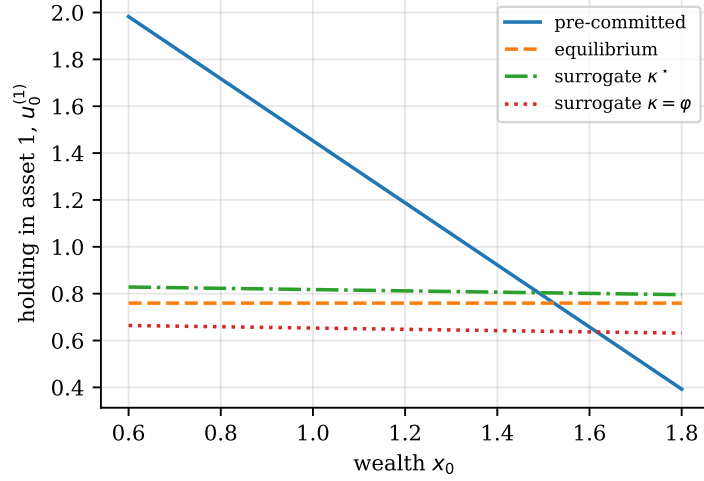


Figure 3: First-period holding in asset one as a function of initial wealth. The equilibrium control is flat (Theorem 5 proof, $A_t^2 = B_t$); the pre-committed control decreases in wealth; the surrogate lies between.

7 The differential Sharpe ratio

The reward class treated so far is the variance-penalised family. Both surveys report the differential Sharpe ratio of Moody and Saffell (2001) as a further standard choice, and it does not obviously belong to that family: it depends on statistics accumulated along the trajectory and so is not a function of wealth alone.

Write ρ_t for the period return and let A_t, B_t be exponentially weighted estimates of its first two moments with decay η ,

$$A_t = A_{t-1} + \eta(\rho_t - A_{t-1}), \quad B_t = B_{t-1} + \eta(\rho_t^2 - B_{t-1}),$$

$$r_t^{\text{DSR}} = \frac{B_{t-1}\Delta A_t - \frac{1}{2}A_{t-1}\Delta B_t}{(B_{t-1} - A_{t-1}^2)^{3/2}}.$$

Proposition 11. *Up to a positive state-dependent scale and an additive constant,*

$$r_t^{\text{DSR}} \propto \rho_t - \kappa_t^{\text{eff}} \rho_t^2, \quad \kappa_t^{\text{eff}} = \frac{A_{t-1}}{2B_{t-1}}.$$

Proof. Substituting the increments, $B\Delta A - \frac{1}{2}A\Delta B = \frac{\eta}{2}(2B\rho - A\rho^2 - AB)$. Dividing by the denominator, which is positive and independent of ρ_t , gives a term linear in ρ_t with coefficient $\eta B(B - A^2)^{-3/2}$, a term in ρ_t^2 with coefficient $-\frac{\eta}{2}A(B - A^2)^{-3/2}$, and a constant. The ratio of the second to the first is $A/(2B)$. $\square$

So the differential Sharpe ratio is a variance-penalised reward after all. The distinction is that its penalty is not chosen. Since A_t and B_t are consistent estimators of the first two moments of the realised return, in the stationary regime

$$\kappa^{\text{eff}} \longrightarrow \frac{\mathbb{E}[\rho]}{2\mathbb{E}[\rho^2]}, \quad (14)$$

a quantity containing no preference parameter at all. Numerically, over a 200-period horizon the estimates converge to the true moments ($A^* = 0.021636$ against $\mathbb{E}[\rho] = 0.021676$; $B^* = 0.000479$ against $\mathbb{E}[\rho^2] = 0.000482$), giving $\kappa^{\text{eff}} = 22.51$. Optimising the differential Sharpe reward directly

over affine policies and then searching the variance-penalised family for the closest member returns $\kappa = 21.56$, a 4% agreement, at a residual policy distance of 0.019.

Reading Theorem 5 backwards then gives an implied risk aversion

$$\varphi^{\text{implied}} = \frac{\kappa^{\text{eff}}}{1 - \nu} = 27.9 \quad (15)$$

for this market. A practitioner selecting this reward believes they are asking for risk-adjusted return; they are in fact asking for a mean-variance criterion at a risk aversion the market sets and nothing in their setup reports. Varying the decay η over $[0.02, 0.4]$ moves it only between 30.4 and 31.0, so the effect is not a hyperparameter artefact.

The differential Sharpe ratio is therefore covered by Theorem 5 rather than excluded from it, with $|\kappa^{\text{eff}} - \varphi(1 - \nu)|$ as the first term of Λ . What is different is that this term cannot be driven to zero by choice.

8 Out of model

Every result so far lives inside the i.i.d. market where the closed forms exist. We evaluate the four policies under processes matched to the same first two moments but violating the model in different ways: Student- t innovations with four degrees of freedom, GARCH-like volatility clustering, and AR(1) serial correlation.

process	pre-committed	equilibrium	surrogate κ^*	surrogate $\kappa = \varphi$
Gaussian	1.4267	1.3236	1.3261	1.3166
Student- t	1.4252	1.3238	1.3263	1.3166
vol. clustering	1.4037	1.3234	1.3258	1.3163
AR(1) momentum	0.9285	1.1886	1.1865	1.2270

Table 5: J under model misspecification. Fat tails and volatility clustering leave the ranking intact; serial correlation reverses it.

Fat tails and volatility clustering change J in the third decimal and leave the ordering untouched, which is expected: they perturb higher moments while preserving independence across periods, and the analysis uses only the first two moments. The results are not artefacts of Gaussianity.

Serial correlation is different and the ordering reverses. The pre-committed policy is worst affected, falling from 1.427 to 0.929: its wealth slope is calibrated to a drift that momentum makes wrong, and by construction it cannot revise. This is a real limitation rather than a caveat. Every closed form here assumes independence across periods, and once that fails the reference policies are no longer the objects they are claimed to be.

8.1 Statistical power of a backtest

Section 6 shows the Sharpe ratio ordering these policies incorrectly. A natural objection is that the difference may be too small to matter. The question is then whether a finite backtest could detect it.

Drawing n independent replications and asking how often the Sharpe comparison selects the policy with the better mean-variance objective:

More data makes the comparison *more* confidently wrong, because it estimates the wrong quantity with increasing precision. This is the sharpest form of the claim: reward-design error is not merely hard to see in a backtest, it is invisible in a way that additional data aggravates.

replications	Pr(Sharpe selects the J -better policy)
10	0.022
50	0.003
200	0.000
1000	0.000
5000	0.000

9 Limitations and open problems

Priority. The distinction between summed per-period mean-variance tradeoffs and the terminal criterion is standard in portfolio choice (Gârleanu and Pedersen, 2013; DeMiguel et al., 2015; Li and Ng, 2000). What we believe to be new is the quantitative map, Theorem 5. Searches of the equilibrium mean-variance and dynamic-trading literatures did not turn it up, and Section 5.1 offers a reason: the correction is $\mathcal{O}(\Delta)$ and vanishes in continuous time, which is where that literature works. This is an argument, not a proof of priority. The author has not conducted a systematic search and this must be done before the claim is made in a submitted version. If $\kappa^* = \varphi(1 - \nu)$ is known, the contribution reduces to the decomposition, the transfer to reinforcement learning, and Section 6.

Rigour. Theorem 9 is established by series expansion with the error terms confirmed numerically rather than bounded. Conjecture 10 is supported by fitted exponents only.

Scope. Returns are i.i.d. with known moments; there are no transaction costs, no short-sale constraints and no leverage limits. Wealth is unbounded below, so the policies here permit ruin; imposing a no-bankruptcy constraint would forfeit the closed forms on which the whole comparison rests.

Dependent returns. Section 8 shows the ordering reversing under AR(1) serial correlation, so the analysis should not be read as applying to markets with predictability. Proposition 2 extends directly to time-varying (m_t, M_t, s_t) , with ν_t replacing ν ; Theorem 5 then requires $s_t = 1$ for all t and $\kappa = \varphi(1 - \nu_t)$ simultaneously, which is generally infeasible for a single κ . Establishing the size of the resulting gap is open, and is the most important of the extensions listed here.

On $\varepsilon_{\text{surr}}$. The quantity $\varepsilon_{\text{surr}}$ is a difference of scalar values and is not sign-definite; at κ^* in our market it is -0.0025 , the surrogate optimum beating the equilibrium policy. Statements of the form “ $\varepsilon_{\text{surr}} = 0$ if and only if” are therefore false and all results here are stated for the policy distance, with the value gap as a consequence.

Reproducibility

All numerical results are produced by the accompanying code. The verification suite runs ten independent checks, including agreement between the closed-form pre-committed policy and an unconstrained numerical maximiser to $2.6\text{e-}12$, and direct testing of the equilibrium condition (4) by attempting to beat it.

A Derivation of the constant in Theorem 9

Set $s = 1 + \epsilon$ and $T = 2$. All quantities are expanded to the order needed for the ϵ^2 coefficient of the squared distance.

Surrogate side. From Proposition 2 with $p_2 = q_2 = 0$: $g_1 = \kappa\epsilon$, so

$$\alpha_1 = -d\epsilon, \quad \beta_1 = \frac{d}{2\kappa}, \quad p_1 = -(1-\nu)\kappa\epsilon^2, \quad 1 + q_1 = 1 + \epsilon(1-\nu) + O(\epsilon^2).$$

Then $g_0 = \kappa\epsilon + O(\epsilon^2)$ and $\kappa - p_1 = \kappa(1 + O(\epsilon^2))$, so

$$\alpha_0 = -d\epsilon + O(\epsilon^2), \quad \beta_0 = \frac{d}{2\kappa}(1 + \epsilon(1-\nu)) + O(\epsilon^2).$$

The wealth slope is thus $-d\epsilon$ at both dates, independent of κ at leading order. This is the source of the irreducible gap.

Equilibrium side. From the proof of Theorem 5, $\alpha_t^{\text{eq}} = 0$, $A_t = s^{T-t}$, $B_t = s^{2(T-t)}$ and $b_t = 2sa_t$, whence $D_{t+1} = s^{2(T-t-1)}(M - mm^\top)$ and

$$\beta_1^{\text{eq}} = \frac{d}{2\varphi(1-\nu)}, \quad \beta_0^{\text{eq}} = \frac{d}{2\varphi s(1-\nu)}.$$

Distance. Write $b^* = (2\varphi(1-\nu))^{-1}$ and parametrise $\kappa = \kappa^*(1 + w\epsilon)$ with $\kappa^* = \varphi(1-\nu)$, so that $(2\kappa)^{-1} = b^*(1 - w\epsilon) + O(\epsilon^2)$. Using (11) with x_0 deterministic and $\mathbb{E}[x_1] = s + \nu\beta_0$, $\mathbb{E}[x_1^2] = s^2 + 2s\nu\beta_0 + \nu\beta_0^2$ at leading order, the squared distance in units of $\|d\|^2$ is

$$(\alpha_0 + \beta_0 - \beta_0^{\text{eq}})^2 + \alpha_1^2 \mathbb{E}[x_1^2] + 2\alpha_1(\beta_1 - \beta_1^{\text{eq}})\mathbb{E}[x_1] + (\beta_1 - \beta_1^{\text{eq}})^2.$$

Expanding and retaining the ϵ^2 coefficient gives a quadratic in w . Minimising over w and simplifying yields

$$\min_w \frac{\|\pi^\kappa - \pi^{\text{eq}}\|^2}{\|d\|^2 \epsilon^2} = \frac{(2-\nu)(1+\nu)}{4\varphi^2(1-\nu)^2},$$

and taking the square root gives the constant of Theorem 9. The algebra was carried out with a computer algebra system and the resulting expression checked against direct numerical minimisation on six markets, agreeing to 0.11% worst case with fitted exponent 1.0000.

Remark 12. The step that does not generalise mechanically to $T \geq 3$ is the last: the number of terms in the ϵ^2 coefficient grows with the horizon, and while w^* remains the minimiser of a quadratic, we have not found a closed form for the resulting expression.

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